```
// ============================================
// EXP-14: TCP SERVER PROGRAM (Windows Winsock)
// ============================================
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <winsock2.h>
#include <ws2tcpip.h>

#pragma comment(lib, "ws2_32.lib")

int main()
{
   WSADATA wsa;
   char buff[512];
   SOCKET SocketFD, connectFD;
   struct sockaddr_in stSockAddr, cli_addr;
   int a;

   // Initialize Winsock
   if (WSAStartup(MAKEWORD(2, 2), &wsa) != 0)
   {
      perror("WSAStartup failed");
      exit(EXIT_FAILURE);
   }

   // Create Socket
   SocketFD = socket(AF_INET, SOCK_STREAM, 0);
   if (SocketFD == INVALID_SOCKET)
   {
      perror("Cannot create socket");
      WSACleanup();
      exit(EXIT_FAILURE);
   }

   // Configure Server Address Structure
   memset(&stSockAddr, 0, sizeof(stSockAddr));
   stSockAddr.sin_family = AF_INET;
   stSockAddr.sin_port = htons(1109);
   stSockAddr.sin_addr.s_addr = INADDR_ANY;

   // Bind Socket to Port
   if (bind(SocketFD, (struct sockaddr*)&stSockAddr, sizeof(stSockAddr)) == SOCKET_ERROR)
   {
      perror("Cannot bind");
      closesocket(SocketFD);
      WSACleanup();
      exit(EXIT_FAILURE);
   }

   // Listen for incoming connections
   listen(SocketFD, 2);
   printf("Server listening on port 1109...\n");

   // Accept client connection
   a = sizeof(cli_addr);
   connectFD = accept(SocketFD, (struct sockaddr*)&cli_addr, &a);
   if (connectFD == INVALID_SOCKET)
   {
      perror("Cannot connect");
      closesocket(SocketFD);
      WSACleanup();
      exit(EXIT_FAILURE);
   }

   // Communication Loop
```

```
    while (1)
    {
        printf("\nReceiving data from client.....");
        memset(buff, 0, sizeof(buff));
        recv(connectFD, buff, 512, 0);

        if (strcmp(buff, "x") == 0)
            break;

        printf("\nData is: %s", buff);

        printf("\nEnter data to be sent: ");
        gets(buff);
        send(connectFD, buff, 512, 0);
    }

    // Cleanup and Close Socket
    closesocket(connectFD);
    closesocket(SocketFD);
    WSACleanup();
    printf("\nConnection closed.\n");
    return EXIT_SUCCESS;
}
// ============================================
// EXP-14: TCP CLIENT PROGRAM (Windows Winsock)
// ============================================

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <winsock2.h>
#include <ws2tcpip.h>

#pragma comment(lib, "ws2_32.lib")

int main()
{
    WSADATA wsa;
    char buff[512];
    SOCKET SocketFD;
    struct sockaddr_in stSockAddr;
    int xyz;

    // Initialize Winsock
    if (WSAStartup(MAKEWORD(2, 2), &wsa) != 0)
    {
        perror("WSAStartup failed");
        exit(EXIT_FAILURE);
    }

    // Create Socket
    SocketFD = socket(AF_INET, SOCK_STREAM, 0);
    if (SocketFD == INVALID_SOCKET)
    {
        perror("Cannot create socket");
        WSACleanup();
        exit(EXIT_FAILURE);
    }

    // Configure Server Address Structure
    memset(&stSockAddr, 0, sizeof(stSockAddr));
    stSockAddr.sin_family = AF_INET;
    stSockAddr.sin_port = htons(1109);
    stSockAddr.sin_addr.s_addr = inet_addr("127.0.0.1");

    // Connect to Server
```

```
    xyz = connect(SocketFD, (struct sockaddr*)&stSockAddr, sizeof(stSockAddr));
    if (xyz == SOCKET_ERROR)
    {
        perror("Cannot connect to server");
        closesocket(SocketFD);
        WSACleanup();
        exit(EXIT_FAILURE);
    }

    // Communication Loop
    while (1)
    {
        printf("\nEnter data to be sent: ");
        gets(buff);
        send(SocketFD, buff, 512, 0);

        if (strcmp(buff, "x") == 0)
            break;

        printf("\nReceiving data from Server.....");
        memset(buff, 0, sizeof(buff));
        recv(SocketFD, buff, 512, 0);

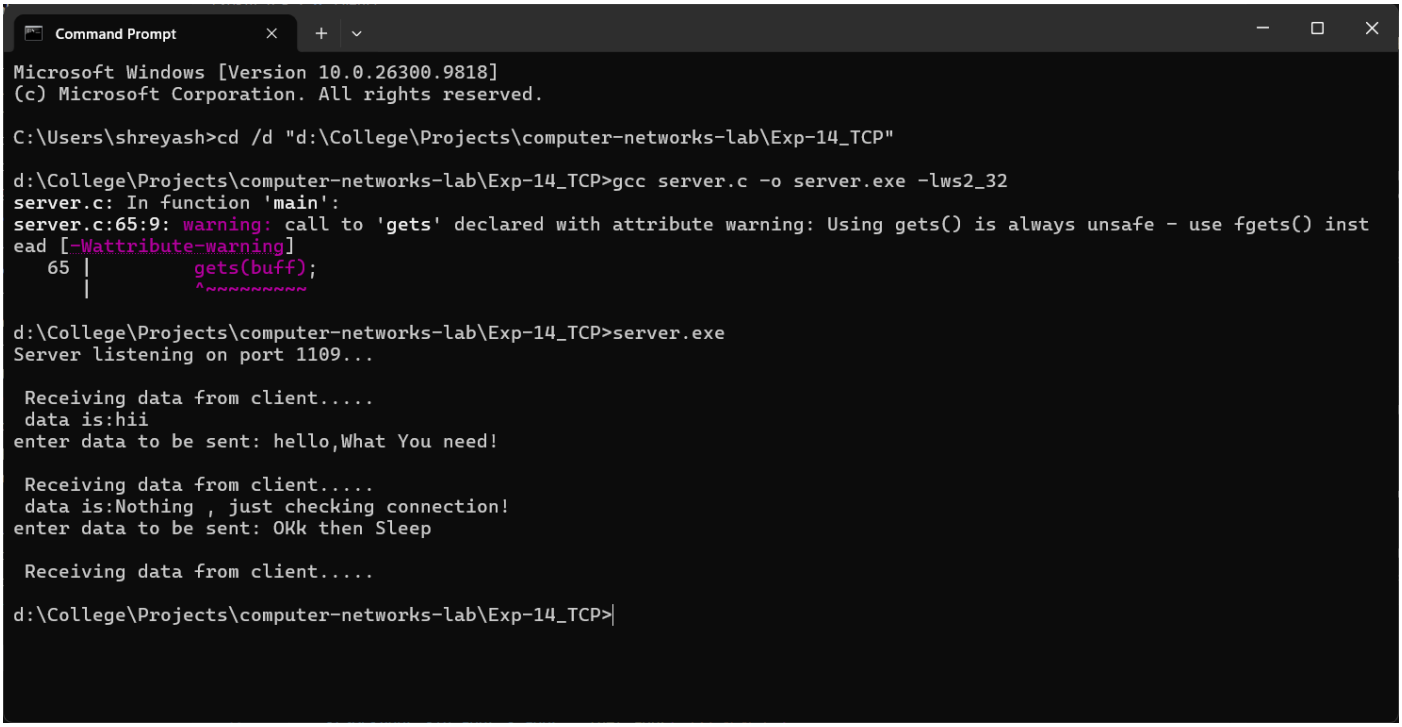
        if (strcmp(buff, "x") == 0)
            break;

        printf("\nData is: %s", buff);
    }

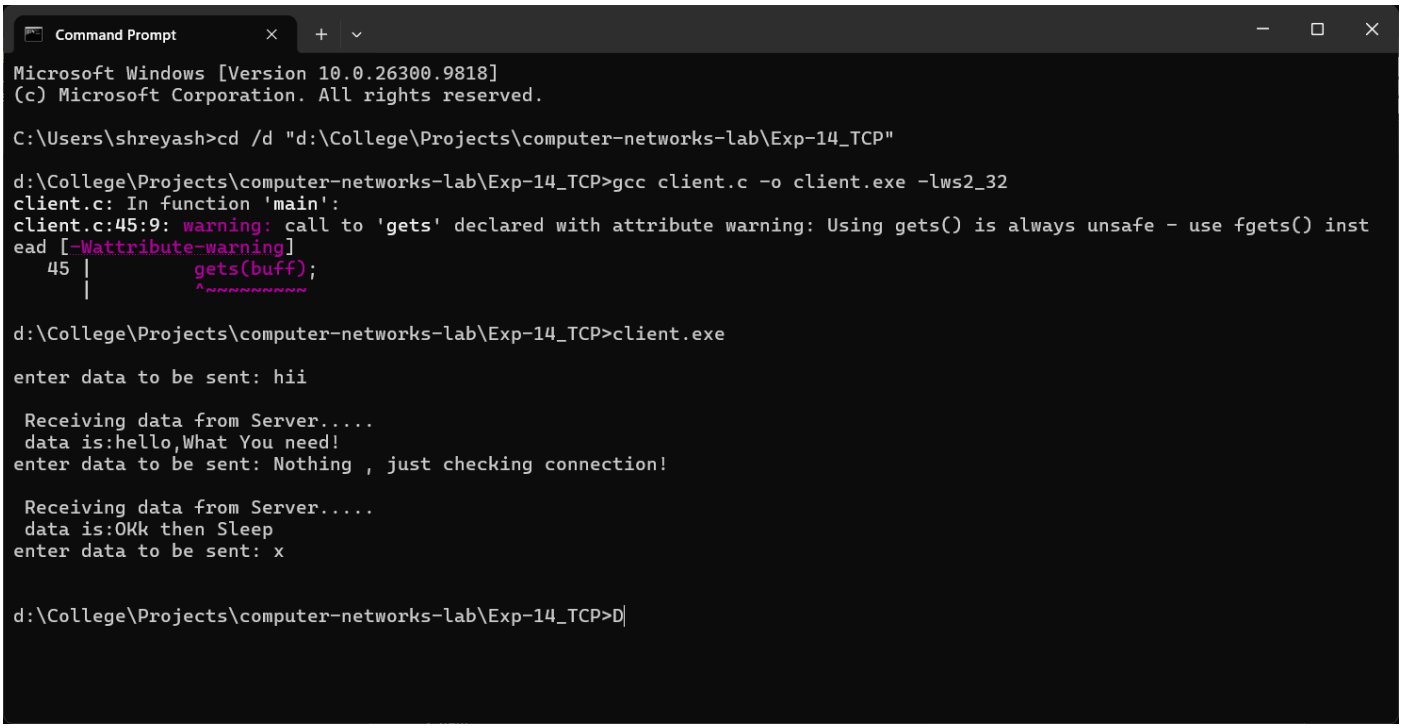
    // Cleanup and Close Socket
    closesocket(SocketFD);
    WSACleanup();
    printf("\nConnection closed.\n");
    return EXIT_SUCCESS;
}
```

OutPut:

Server:



Client:


Command Prompt
Microsoft Windows [Version 10.0.26300.9818]
(c) Microsoft Corporation. All rights reserved.
C:\Users\shreyash>cd /d "d:\College\Projects\computer-networks-lab\Exp-14_TCP"
d:\College\Projects\computer-networks-lab\Exp-14_TCP>gcc client.c -o client.exe -lws2_32
client.c: In function 'main':
client.c:45:9: warning: call to 'gets' declared with attribute warning: Using gets() is always unsafe - use fgets() inst
ead [-Wattribute-warning]
45 | gets(buff);
| ^~~~~~~~~
d:\College\Projects\computer-networks-lab\Exp-14_TCP>client.exe
enter data to be sent: hii
Receiving data from Server.....
data is:hello,What You need!
enter data to be sent: Nothing , just checking connection!
Receiving data from Server.....
data is:OKk then Sleep
enter data to be sent: x
d:\College\Projects\computer-networks-lab\Exp-14_TCP>D